

Applicant: Irfana Aslam

Programme: Erasmus Mundus Joint Master – Image Processing and Computer Vision with Artificial Intelligence (IPCVAI)

Dear Members of the Selection Committee,

I am writing to express my strong motivation to join the Erasmus Mundus Joint Master's programme in Image Processing and Computer Vision with Artificial Intelligence (IPCVAI). With a Bachelor's degree in **Bioinformatics** and a Master's degree in **Computer Science**, hands-on experience in deep learning and computer vision, and a peer-reviewed research publication, I am eager to contribute to and benefit from this interdisciplinary programme.

I earned my BSc in Bioinformatics (CGPA 2.73) and MSc in Computer Science (CGPA 3.25) from COMSATS University Islamabad. My academic journey provided a solid foundation in programming, algorithms, mathematics, and AI, complemented by specialized coursework and projects in **image processing, computer vision, and deep learning**. I am proficient in Python, TensorFlow, OpenCV, and convolutional neural networks, enabling me to tackle complex visual data challenges.

My research includes designing deep learning models for **fabric design defect detection**, focusing on fine-grained feature extraction, localization, and classification. This experience strengthened my skills in dataset preparation, model optimization, and performance evaluation. Additionally, I have **authored the peer-reviewed paper:**

"Design and Assessment of a Novel In Silico Approach for Developing a Next-Generation Multi-Epitope Universal Vaccine Targeting Coronaviruses"

demonstrating my ability to conduct high-impact, interdisciplinary research.

The IPCVAI programme aligns perfectly with my academic and career goals. Its **interdisciplinary curriculum**, international mobility, and emphasis on applied research in **medical imaging and intelligent visual systems** will allow me to deepen my expertise, collaborate with top researchers, and develop AI solutions with tangible societal impact.

Looking ahead, I aspire to pursue doctoral research and contribute to **robust, explainable, and impactful AI systems** in healthcare and scientific applications. I am confident that my academic background, research accomplishments, and passion for AI and computer vision make me a strong candidate for IPCVAI and a valuable addition to its cohort.

Thank you for considering my application. I would be honored to join IPCVAI and contribute to its vibrant academic and research community.

Yours sincerely,

Irfana Aslam